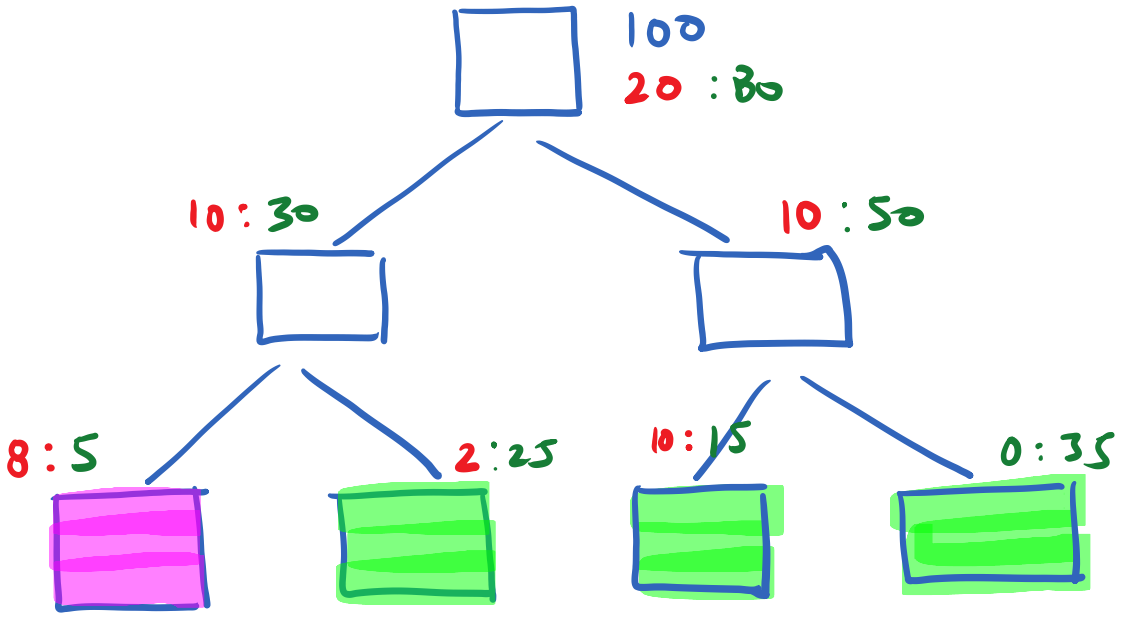
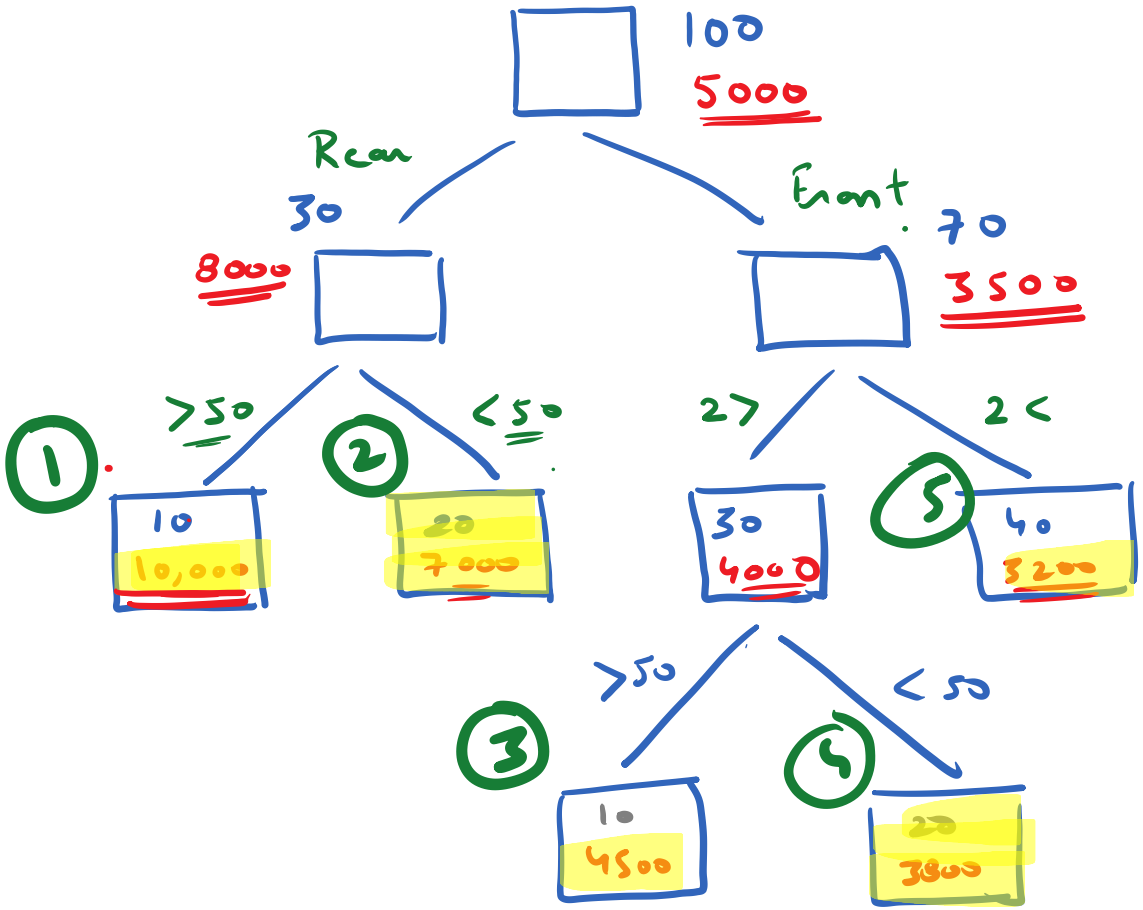
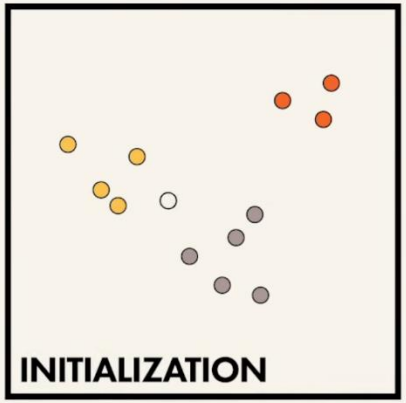
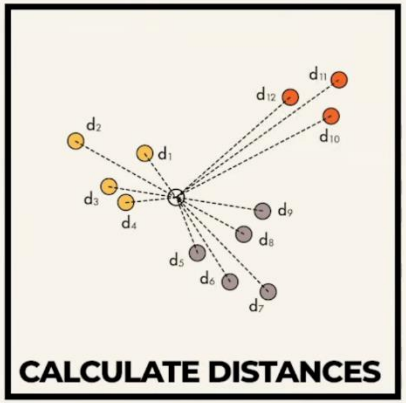
Machine Learning-7

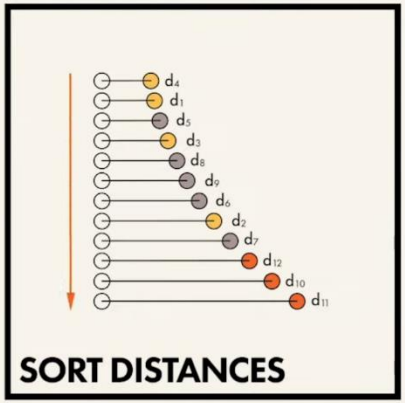

100
20 : 80
10 : 30
10 : 50
8 : 5
2 : 25
10 : 15
0 : 35


100
5000
Rear
30
8000
Front
70
3500
>50
<50
2>
2<
1
2
5
10
10,000
20
7000
30
4000
40
3200
>50
<50
3
4
10
4500
20
3800

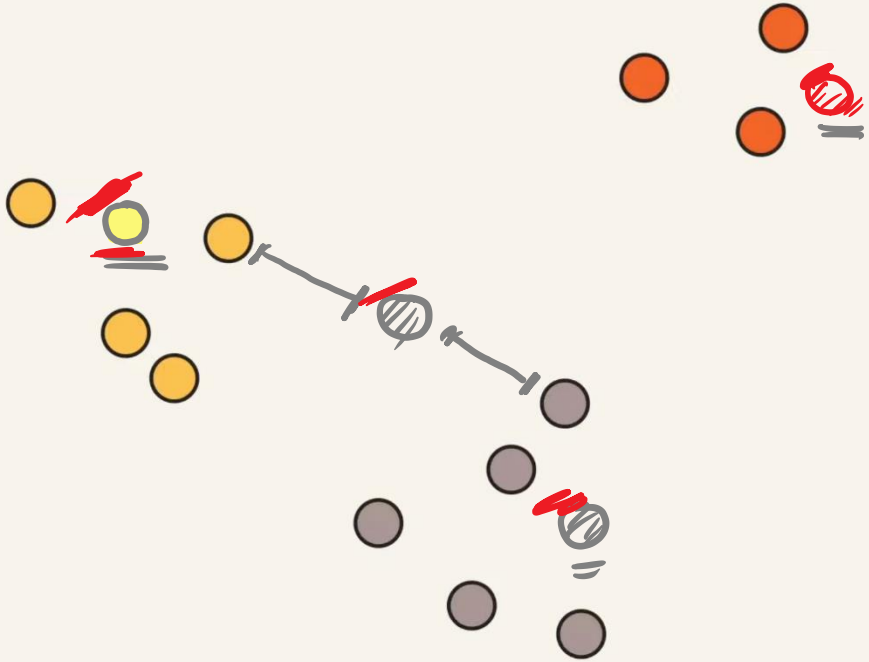
K NEAREST NEIGHBORS







KNN = "K" Nearest Neighbour

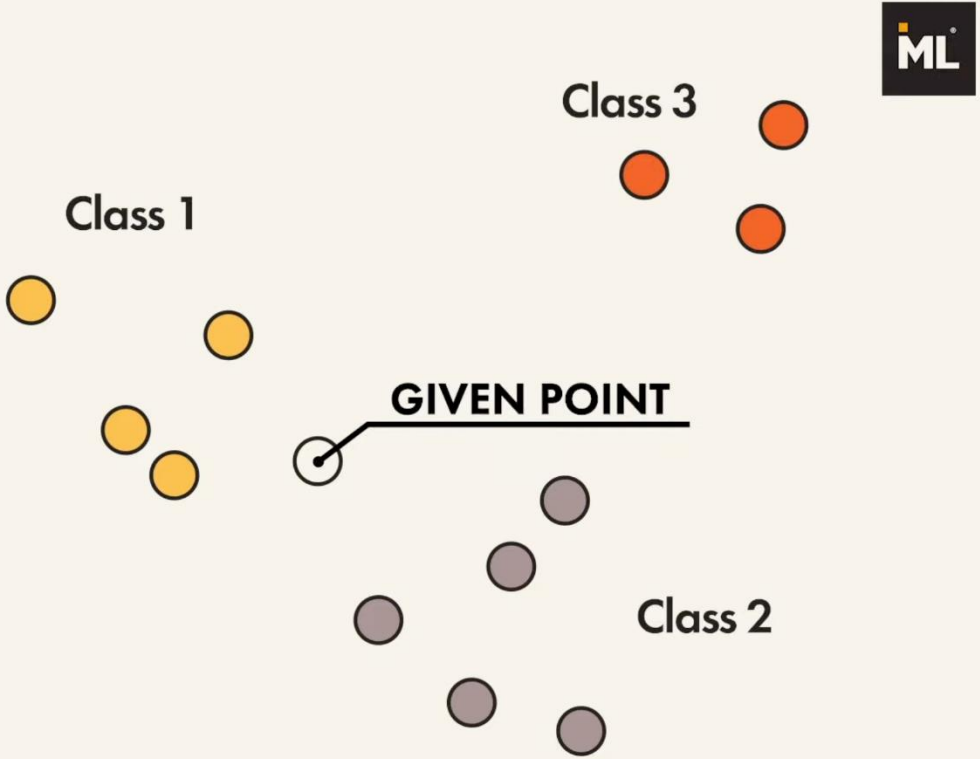


STEP 1

Initialization

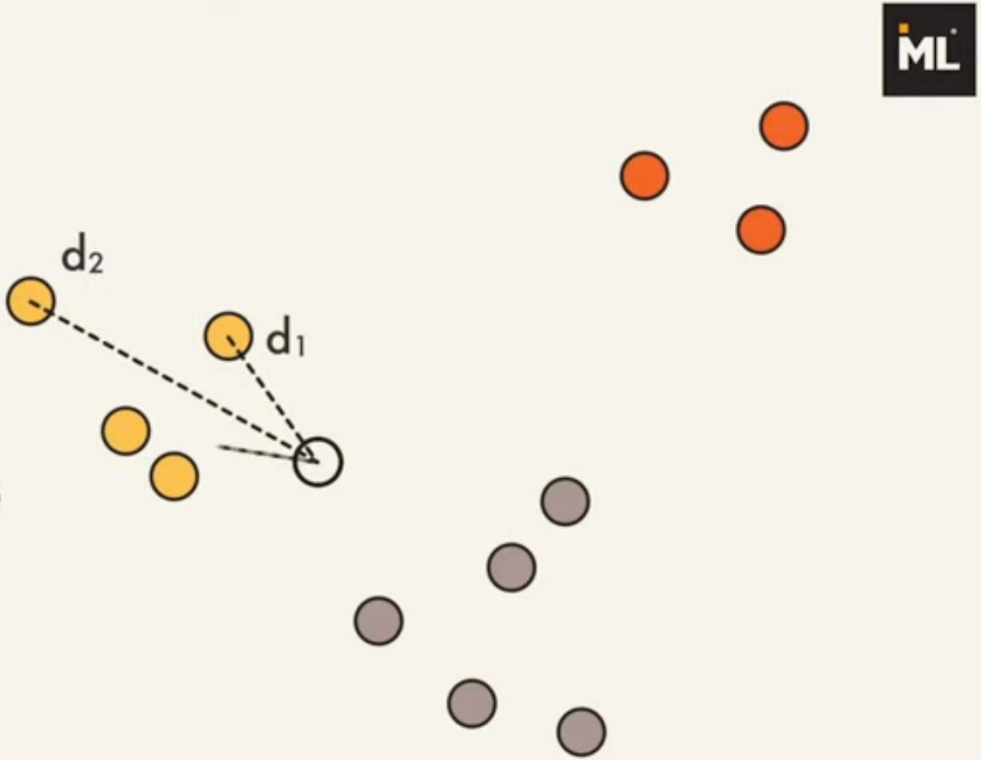
STEP 1

Initialization



STEP 2

Calculate Distance



STEP 2

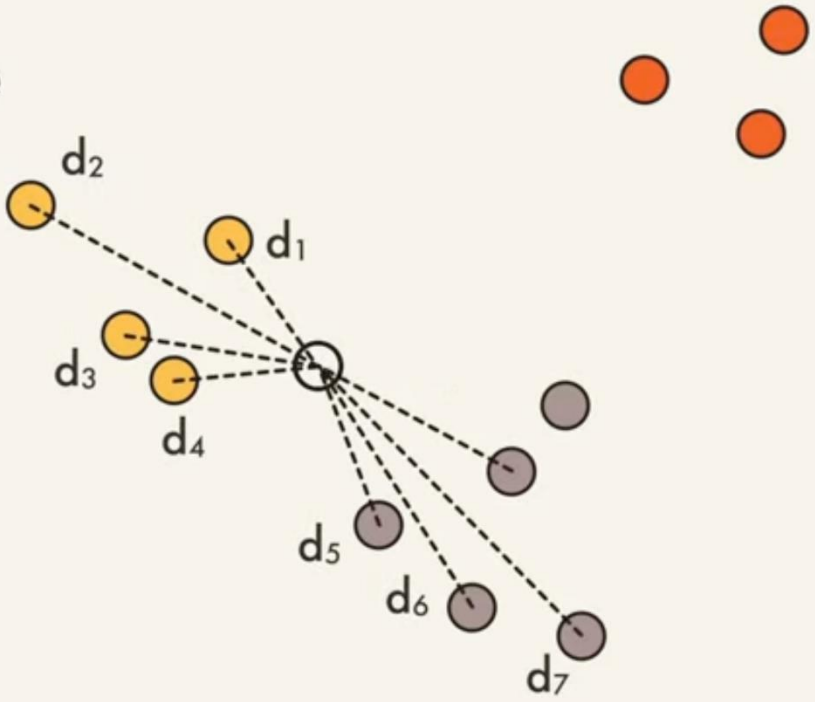
Calculate Distance

Euclidean:

$$d(x,y) = \sqrt{\sum_{i=1}^{m} (x_i - y_i)^2}$$

Manhattan/city - block:

$$d(x,y) = \sum_{i=1}^{m} |x_i - y_i|$$



$B(5, 2)$

$A(2, 0)$

$$AB = \sqrt{(2-5)^2 + (0-2)^2}$$

STEP 2

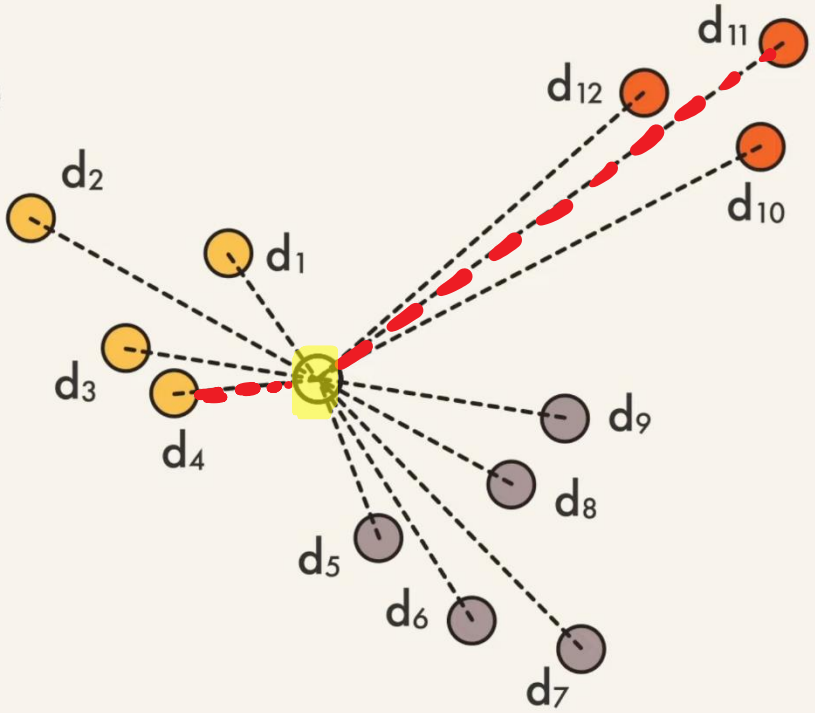
Calculate Distance

Euclidean:

$$d(x,y) = \sqrt{\sum_{i=1}^{m} (x_i - y_i)^2}$$

Manhattan/city - block:

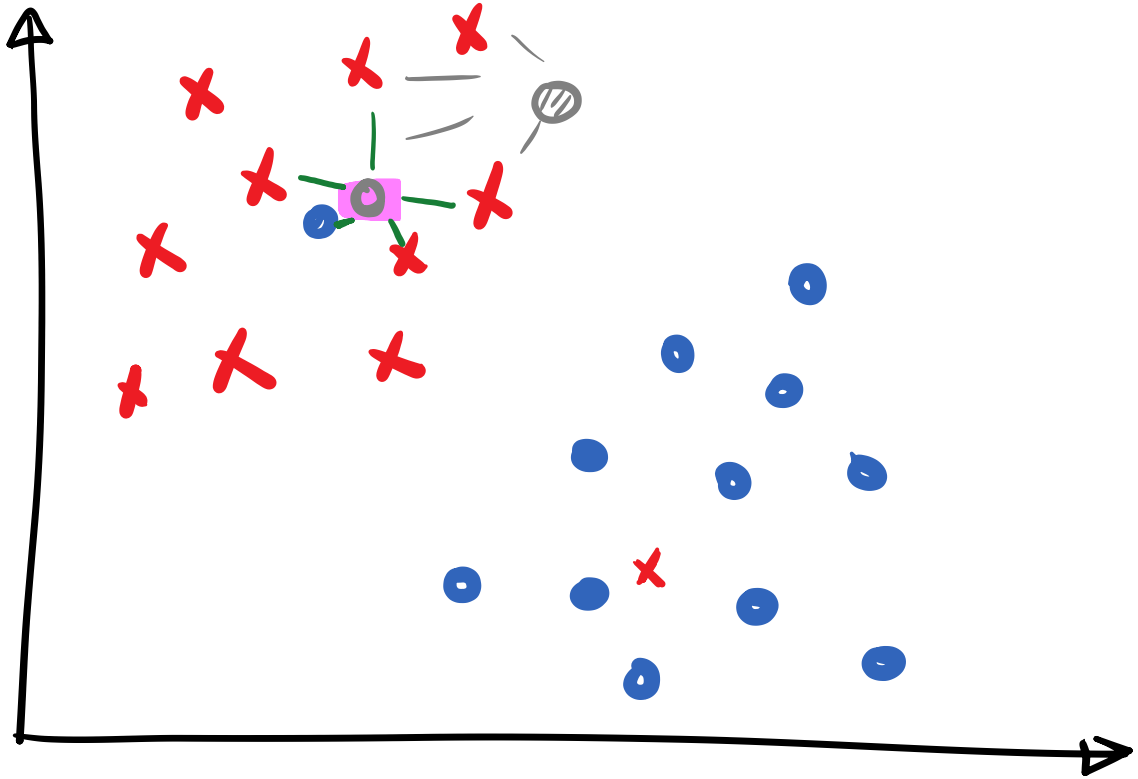
$$d(x,y) = \sum_{i=1}^{m} |x_i - y_i|$$

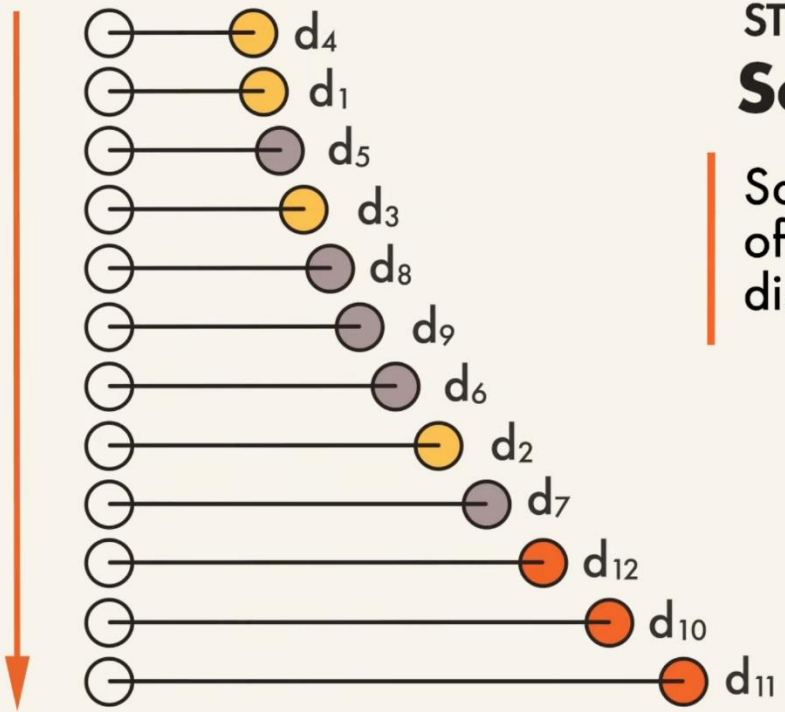


$B(3, 1, 2)$

$A(2, 0, 1)$

$$AB = \sqrt{(2-3)^2 + (0-1)^2 + (1-2)^2}$$

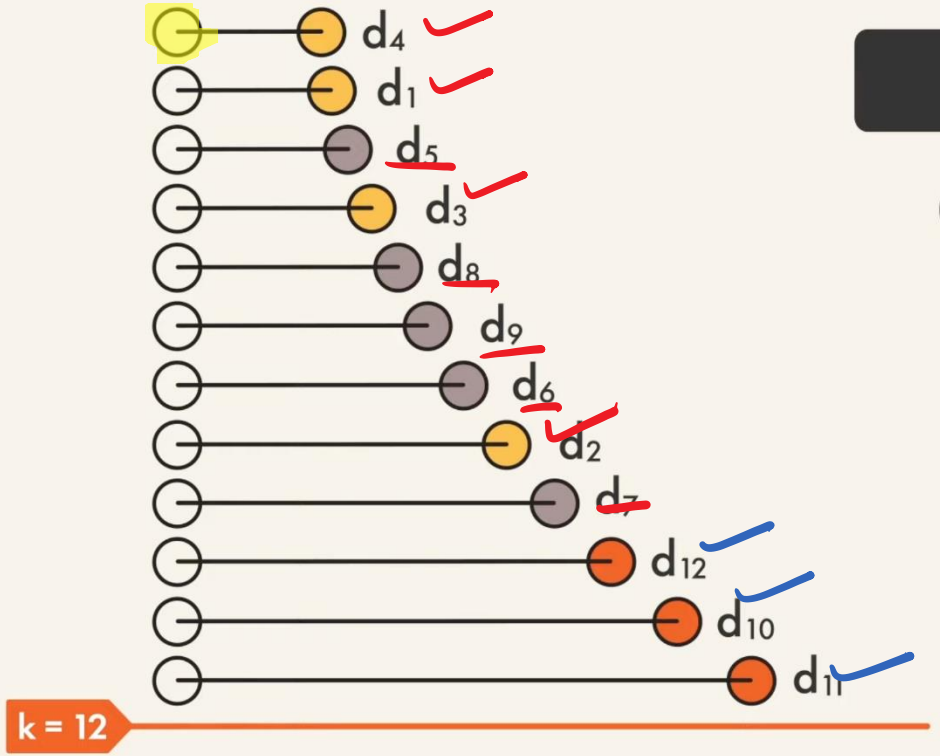




STEP 3

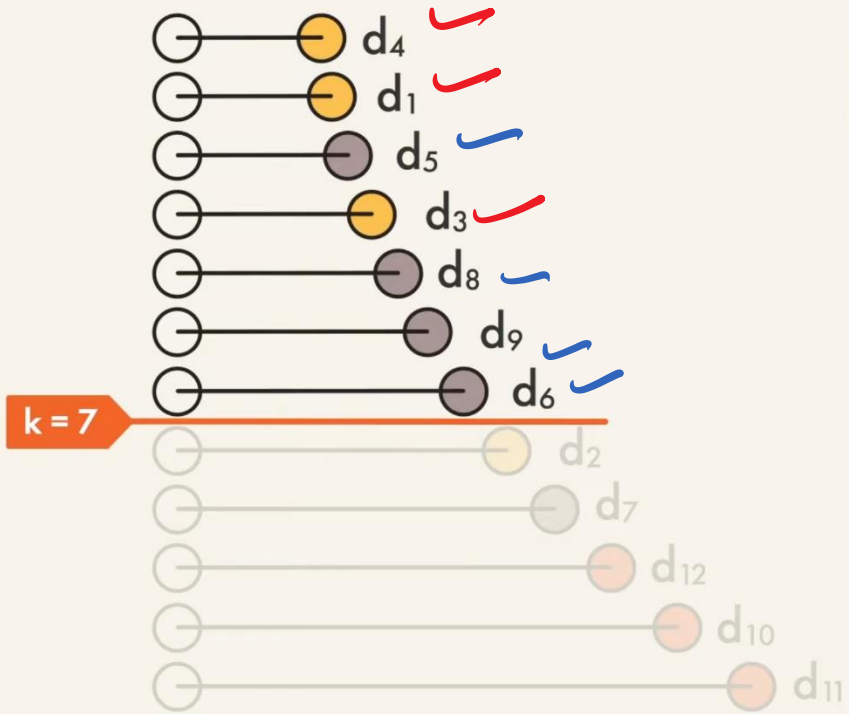
Sort Distance

Sort the nearest neighbors of the given point by the distances in increasing order

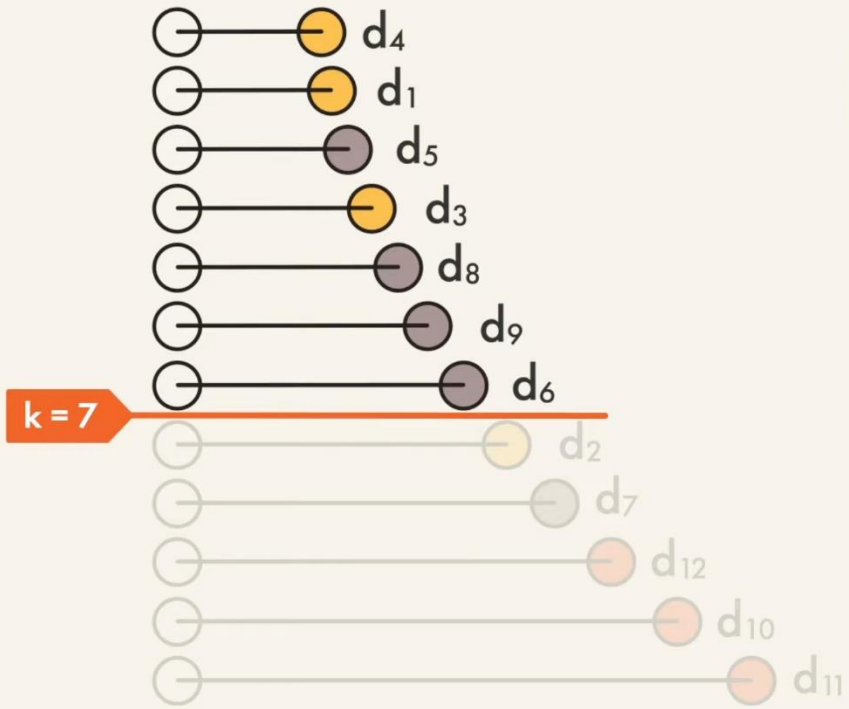


Classification

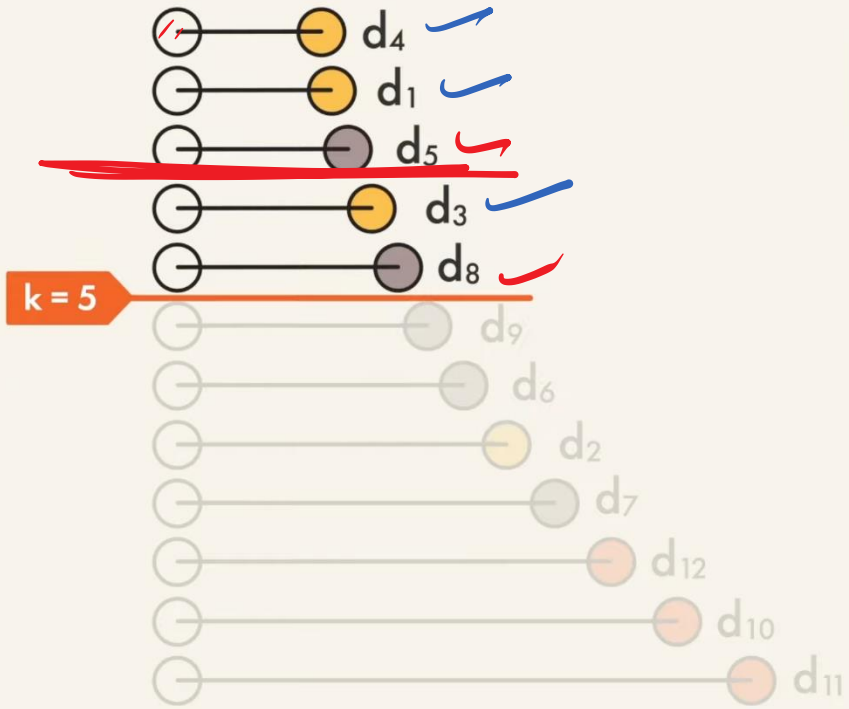
4 5 3



Classification

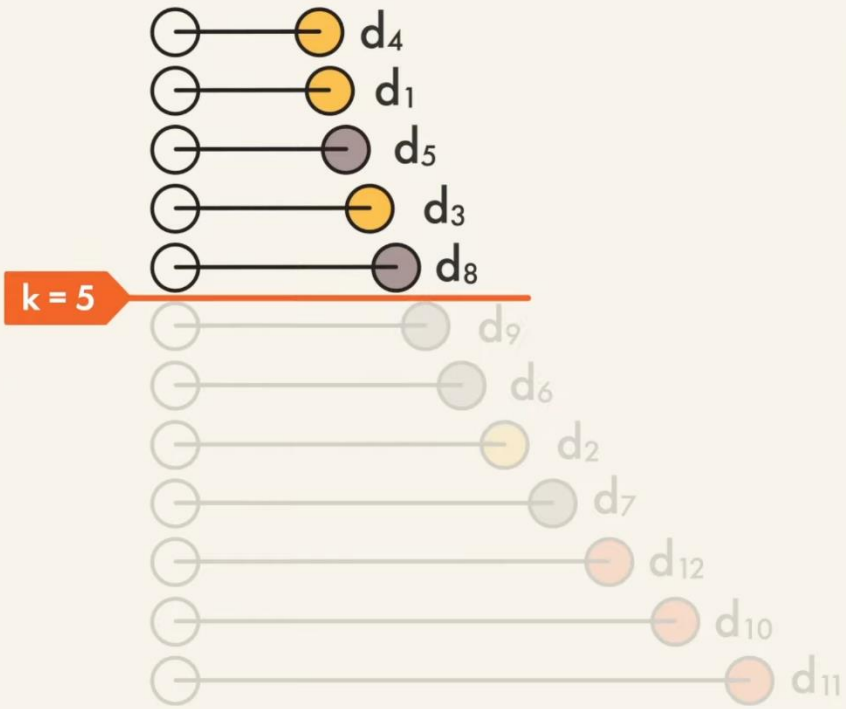


Classification



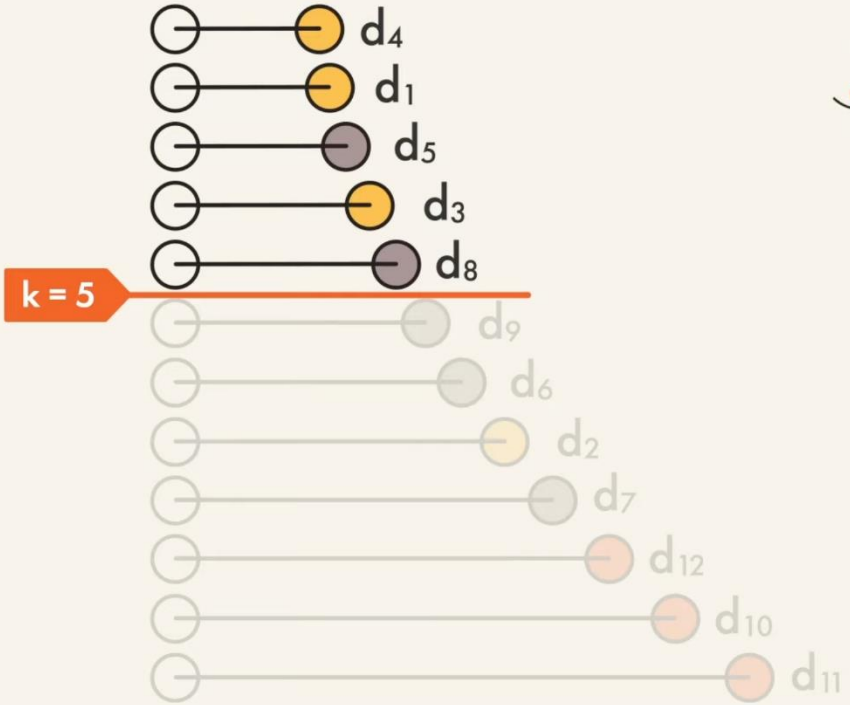
Classification

3 2 0

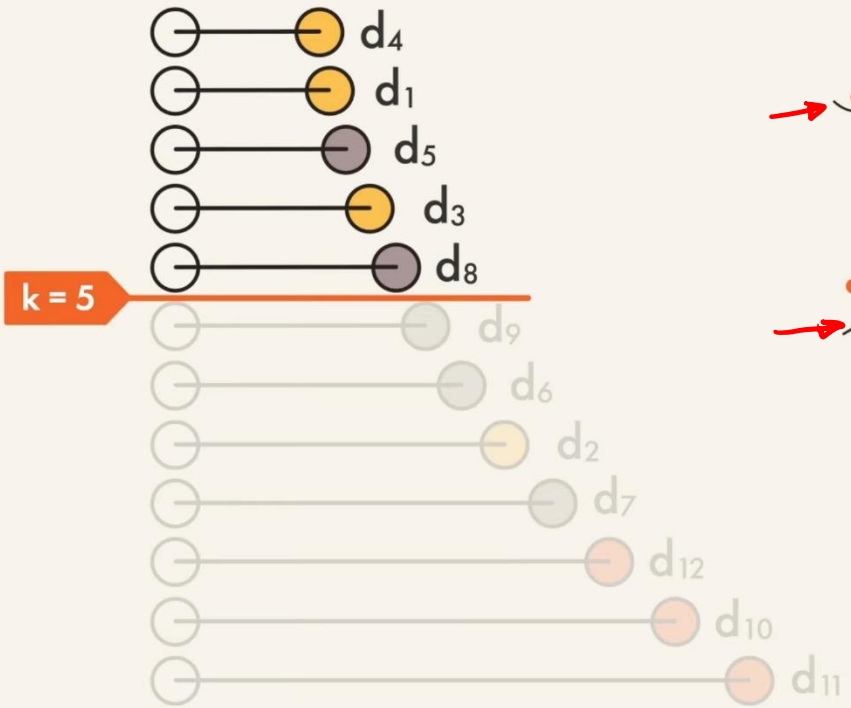


Classification

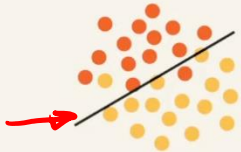
3 2 0



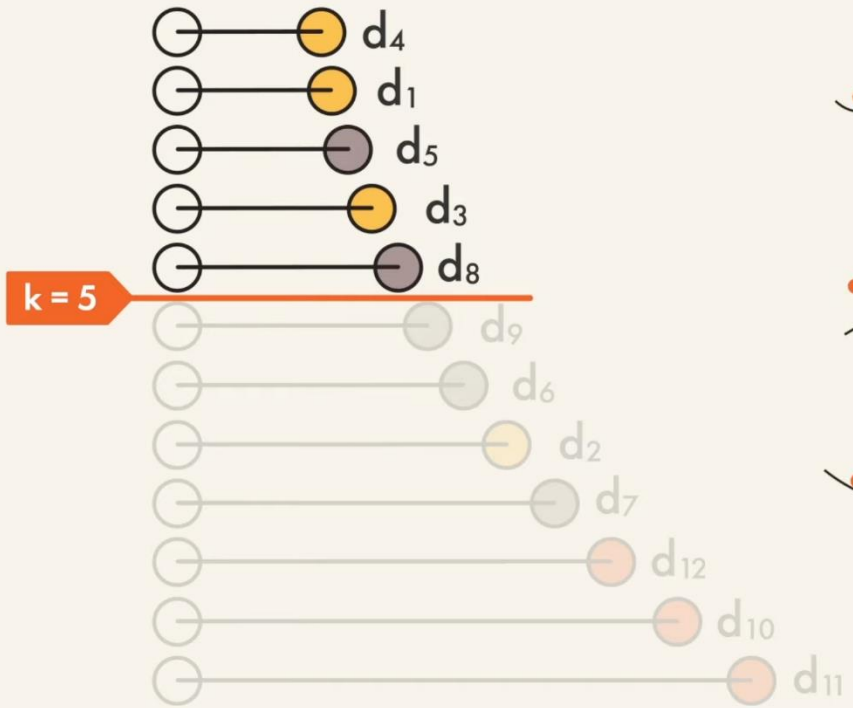
Small K
Low Bias; High Variance
Overfitting



Small K
Low Bias; High Variance
Overfitting



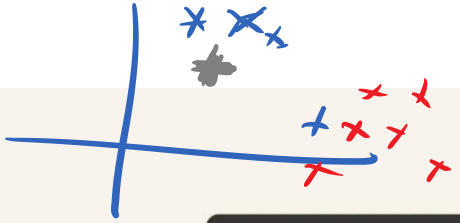
Large K
High Bias; Low Variance
Underfitting

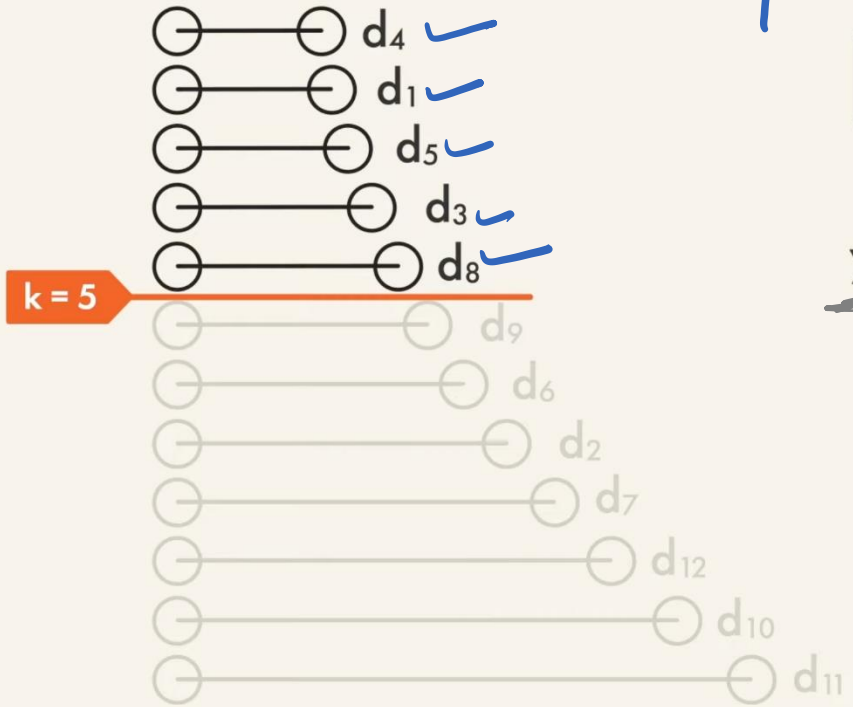


Small K
Low Bias; High Variance
Overfitting

Large K
High Bias; Low Variance
Underfitting

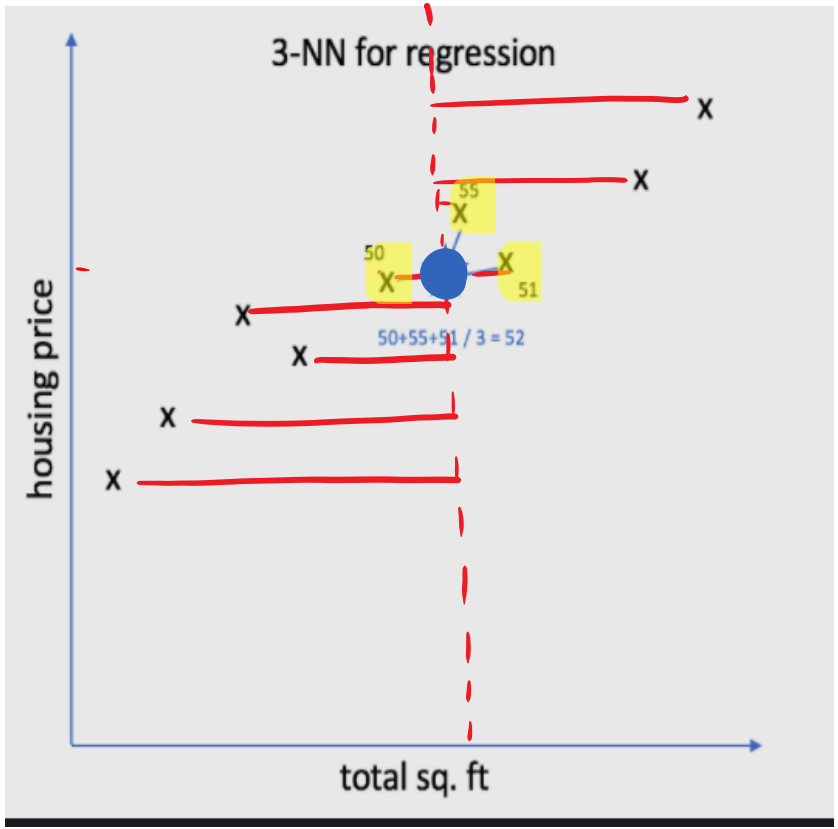
Best K value controls the balance between overfitting and underfitting





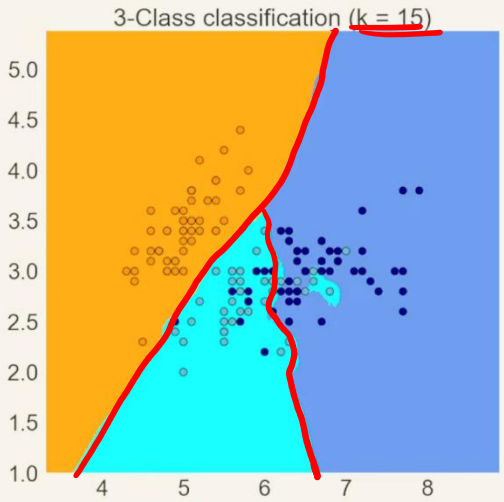
Regression

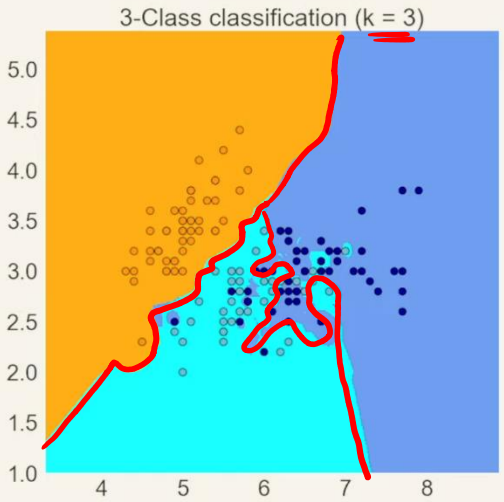
$$y = \frac{y_4 + y_1 + y_5 + y_3 + y_8}{k}$$

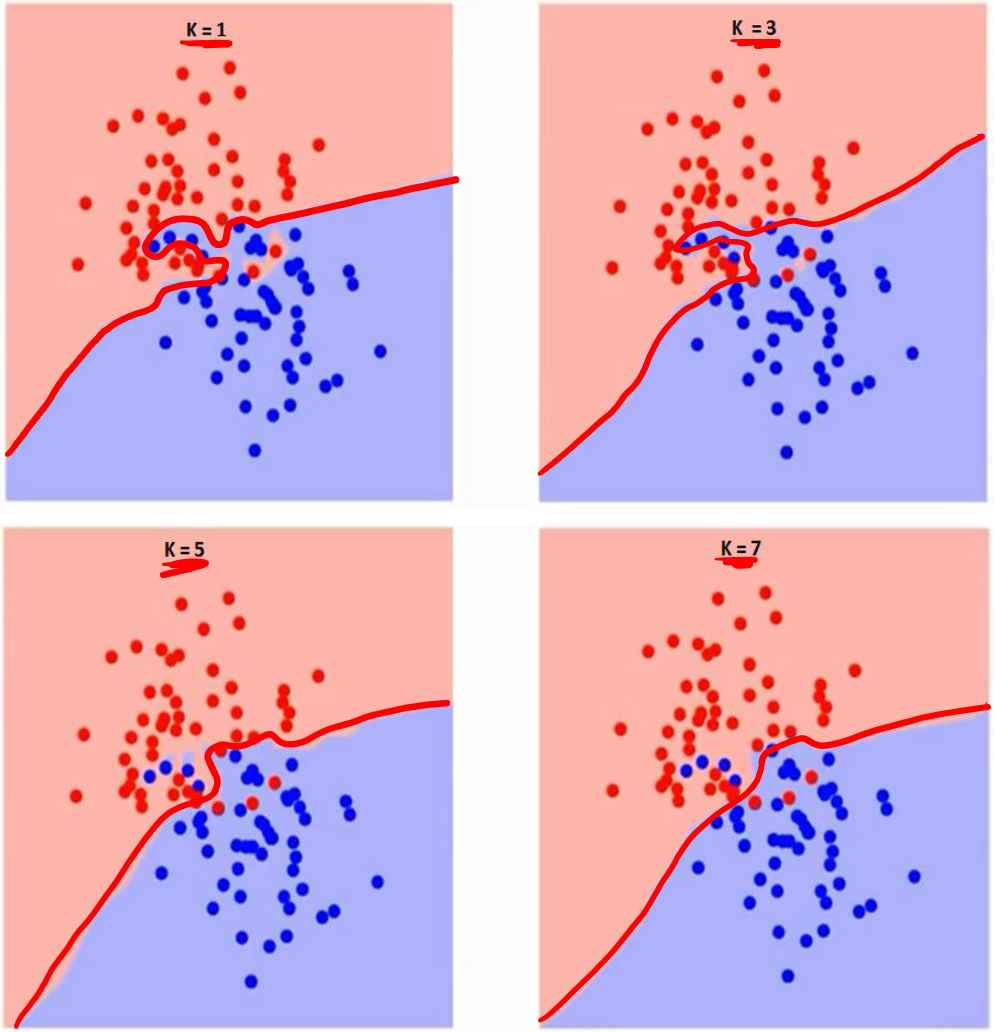

3-NN for regression
housing price
total sq. ft
55
50
51
50+55+51 / 3 = 52

CODE

VISUAL


3-Class classification (k = 15)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
4
5
6
7
8


3-Class classification (k = 3)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
4
5
6
7
8


K = 1
K = 3
K = 5
K = 7

G

B 1 → 10,000

→ 175 cm

B 2 → 10,100

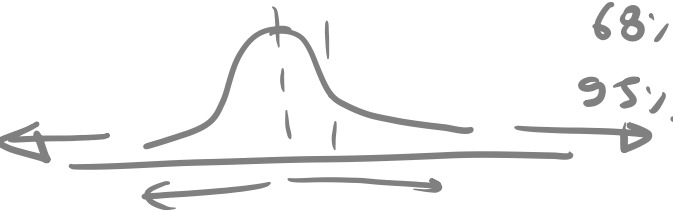
→ 155 cm

B 3 → 10,300

→ 172 cm

$B_1B_2 = \sqrt{(100)^2 + (20)}$

$B_1B_3 = \sqrt{(300)^2 + (5)^2}$



68% → ±σ

95% → ±2σ

+ 99.7% → ±3σ

Normalization = $\dfrac{x_i - x_{min}}{x_{max} - x_{min}}$ ⟹ $[0, 1]$

Standardization = $\dfrac{x_i - \mu}{\sigma} = z$ ⟹ $[-\infty, \infty]$

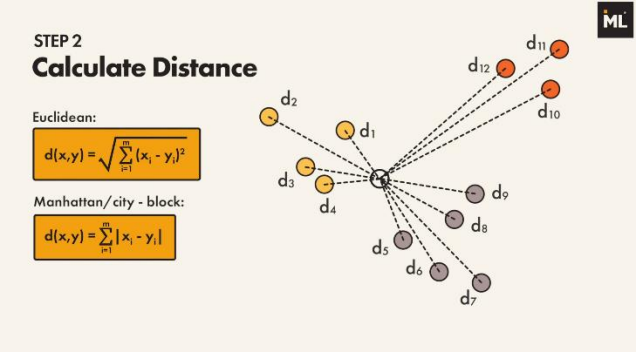
$[-3, 3]$

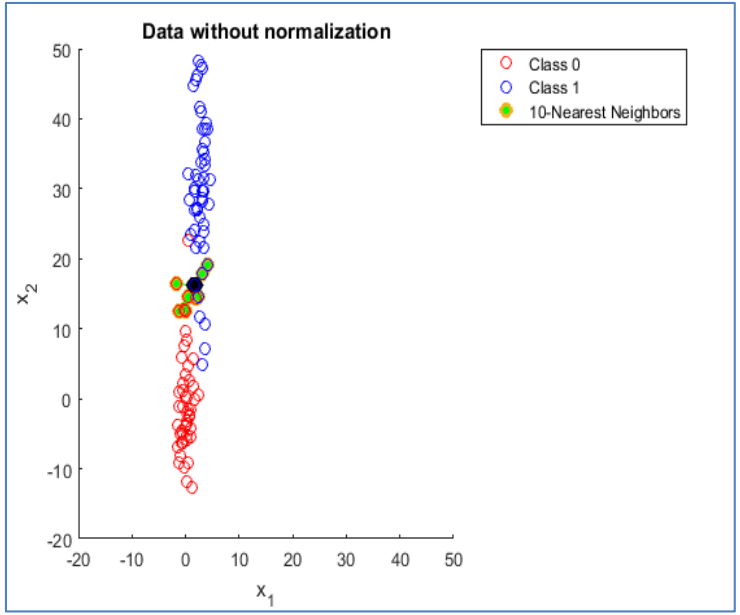
(2, 1)

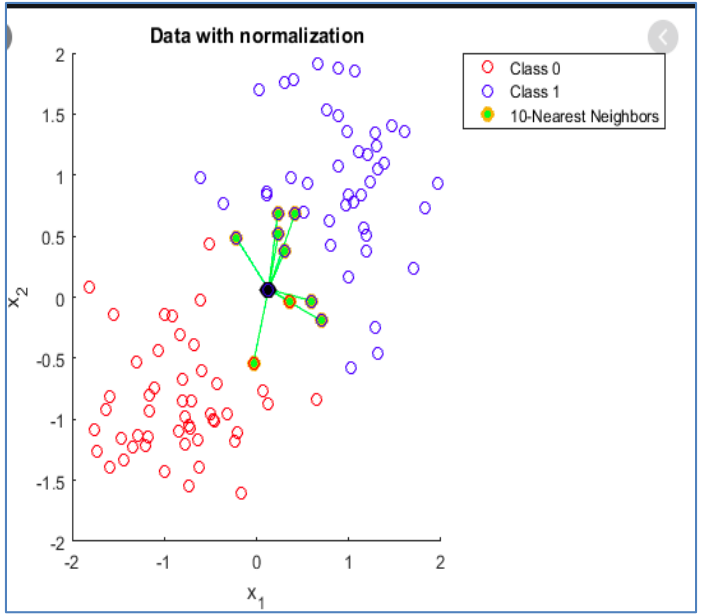
(0.3, 0.1)

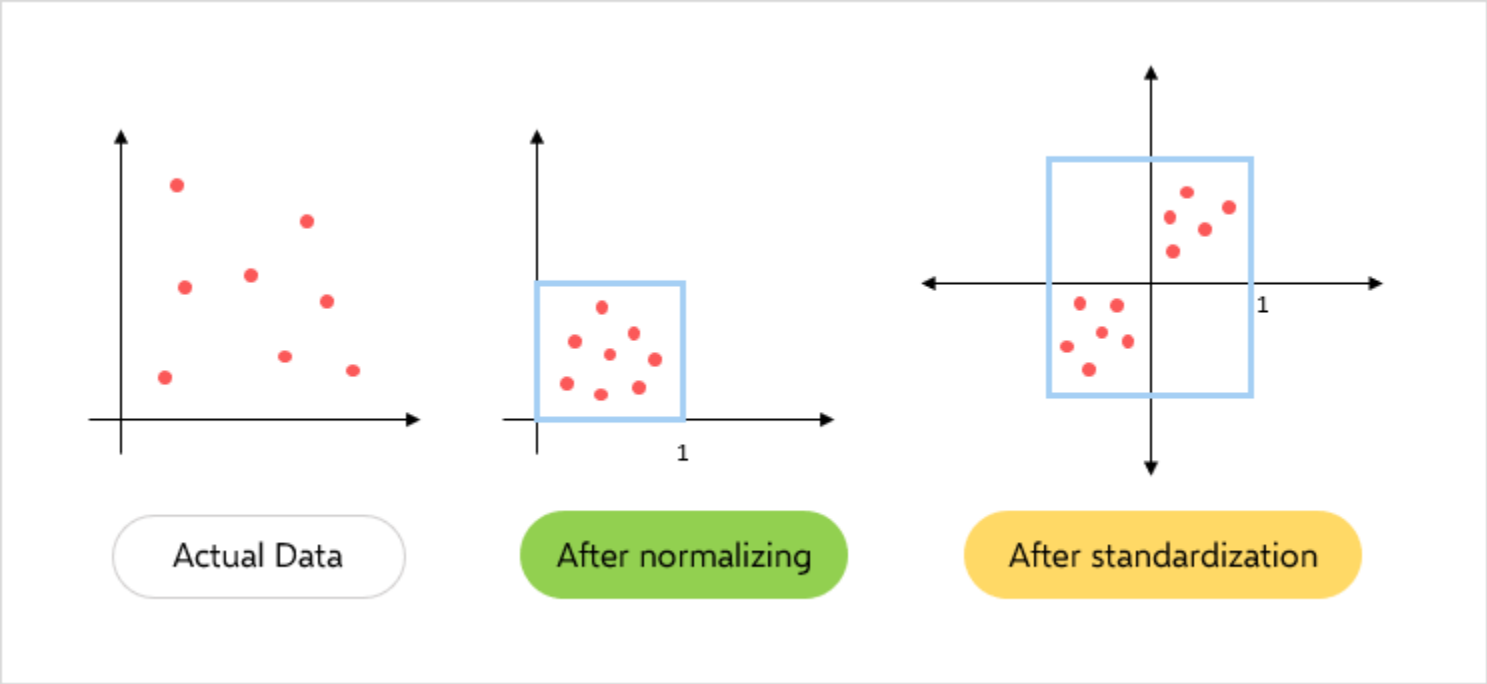
(5, 3)

(0.7, 0.25)


STEP 2
Calculate Distance
Euclidean:
$d(x,y) = \sqrt{\sum_{i=1}^{m} (x_i - y_i)^2}$
Manhattan/city - block:
$d(x,y) = \sum_{i=1}^{m} |x_i - y_i|$
d_1
d_2
d_3
d_4
d_5
d_6
d_7
d_8
d_9
d_{10}
d_{11}
d_{12}


Data without normalization
Class 0
Class 1
10-Nearest Neighbors
x_1
x_2


Data with normalization
Class 0
Class 1
10-Nearest Neighbors
x_1
x_2

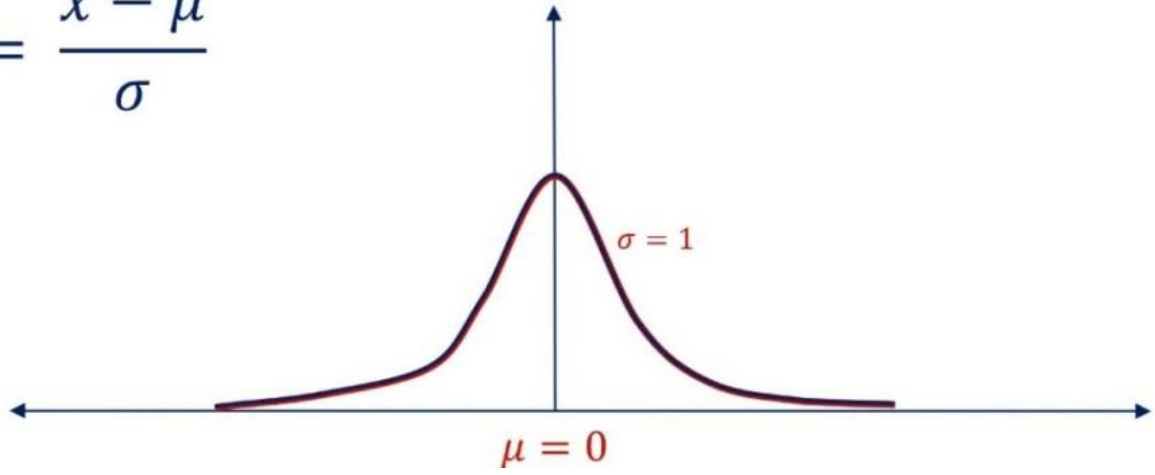

1
1
Actual Data
After normalizing
After standardization

Normalization

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Standardization

$$z = \frac{x - \mu}{\sigma}$$



S.No	Normalization	Standardization
1	Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
2	It is really affected by outliers.	It is much less affected by outliers.
3	It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
4	It is a often called as Scaling Normalization	It is a often called as Z-Score Normalization
5	Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for standardization.

THANK YOU